NATIONAL BIODIVERSITY FUTURE CENTER

NON-INVASIVE APPROACHES FOR 3D RECONSTRUCTION OF TREE ROOT SYSTEMS IN URBAN ENVIRONMENTS

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INTRODUCTION

In the botanical field, to study the morphology of the root system of plants, destructive techniques based on the excavation and cleaning of the roots using the Air Spade are still used; these techniques are very invasive and harmful to the roots and not always applicable, especially in urban environments.

To preserve the health and shape of the root system, it is possible to adopt non-destructive geophysical techniques, such as *Ground Penetrating Radar* (GPR) and *Electrical Resistivity Tomography* (ERT), which allow scanning the ground by acquiring information about the plant roots without having to touch them.

The aim of this research work is to reconstruct a three-dimensional model of the root system of trees, based on the images acquired with the GPR and the ERT.

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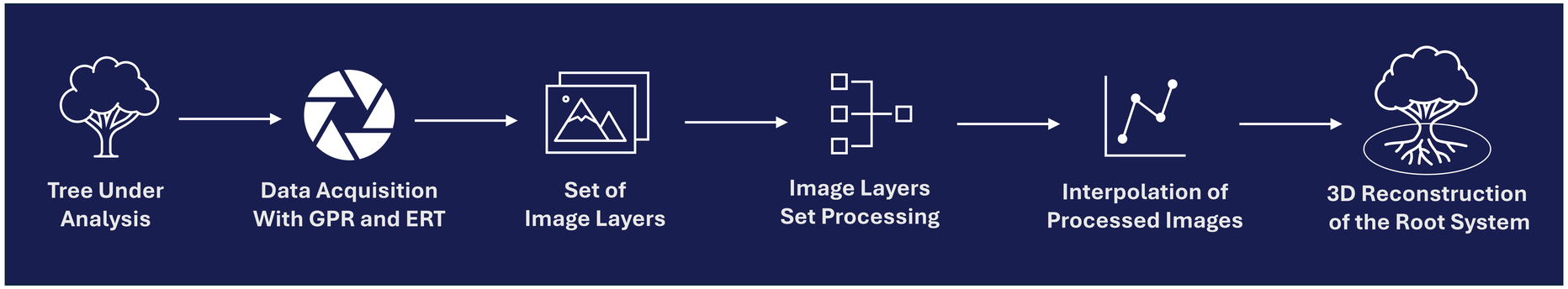

***Figure** 1. Workflow of the proposed method for the 3D reconstruction of the root system.*

METHODOLOGY

We propose a method for the automatic reconstruction of a 3D model of a root system. Figure 1 shows the workflow related to the proposed method.

We start from the acquisition of data relating to the root system, performing two acquisitions: the first with the *GPR* and the second with the *ERT*.
We thus obtain two datasets composed of images of the various depth levels of our scans. We apply a series of filters to the subsoil maps obtained with the scans to clean up the noise present in them and then isolate only the areas of our interest.
Once all the images have been processed, an interpolation algorithm is applied to merge all the layers of each scan, reconstructing our 3D model.

Here are two sample images. Figure 2 shows the set of sample images, where blue represents the background and red represents the root slice for each layer. By applying filters that allow us to select only the area of our interest (i.e.: red color) and the interpolation algorithm, it is possible to obtain the 3D model shown in Figure 3.

To verify the accuracy of the 3D model obtained with our tool, the plant under examination must be excavated, cleaned and scanned with the laser scanner. The verification consists in comparing the 3D model of the root system with that produced by the point cloud of the laser scanner.

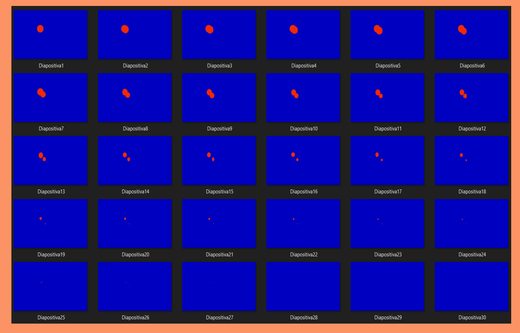

***Figure 2**. Set of 30 sample images.*

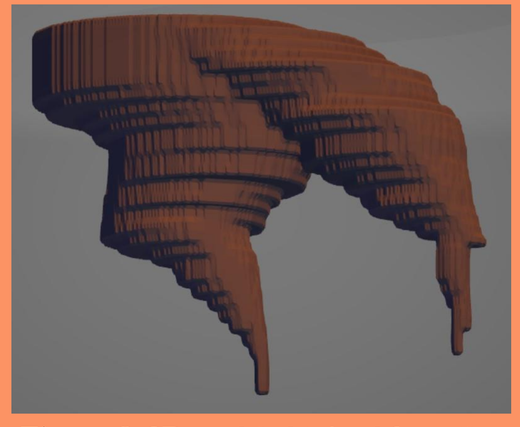

***Figure 3**. 3D model obtained from the sample images.*